Lakshmi Sindhu Pulugundla
Chicago, IL | (312) 776-7472 | sindhu.pl@outlook.com | LinkedIn: /in/lakshmi-sindhu-p/ | GitHub: /Lakshmi-Sindhu-P

EDUCATION

Illinois Institute of Technology - Chicago, IL | *Master of Data Science* — *Graduation Date: Jul 2025*

Presidency University - Bangalore, India | *Bachelor of Technology, Computer Engineering & Data Science* — *Graduation Date: Jul 2023*

WORK EXPERIENCE

Effexoft Inc — **Remote**
Data Analyst — *Aug 2025 – Present*

- Developing FlowLens, an end-to-end supply chain analytics platform built on 547K+ Effecta stock ledger records — engineering ELT pipelines from Salesforce exports into DuckDB and Snowflake using SQL, Python, dbt-core, and Tableau dashboards.
- Designed 35+ analytical features across inventory, vendor, and procurement dimensions; enforced data quality contracts using Pandera validation schemas to monitor schema consistency and quarantine anomalous records before analysis
- Supported Agile delivery for a cross-functional team using Jira and Microsoft Teams Planner; maintained PRD/SRD documentation and transformation workflow references to improve stakeholder alignment, traceability, and analytics pipeline maintainability.

TekLink International, LLC — **Warrenville, IL**
Associate Data Analytics Intern — *Feb 2025 – Aug 2025*

- Designed and deployed a business intelligence reporting platform for healthcare equipment distribution, building Star Schema data models, ETL pipelines, and currency normalization across six data sources via Azure Data Factory, Azure SQL Database, ADLS & Power BI.
- Engineered a fraud detection pipeline for 5.8M+ banking transactions implementing medallion architecture (Bronze/Silver/Gold), open-source AutoML (H2O/FLAML), Unity Catalog data governance, and exploratory data analysis across 183K labeled fraud cases via Delta Live Tables, SparkSQL & Databricks Lakehouse Platform.

Illinois Institute of Technology — **Chicago, IL**
Graduate Student Assistant — *Sep 2024 – May 2025*

- Ensured 100% data accuracy compliance for student enrollment datasets by validating and reconciling records across multiple systems
- Streamlined data entry and verification workflows, reducing processing errors, scheduling conflicts and improving operational efficiency
- Managed reporting & administrative datasets for 1,000+ students, improving data accessibility and reducing turnaround time

AlertFusion Ltd. — **United Kingdom**
Data Science Intern — *Jan 2023 – May 2023*

- Conducted technical feasibility assessment for a cybersecurity alert recommendation system; evaluated ML.NET and Big Data architectures against 7-8 dimensional nested dynamic data, informing solution design decisions

PROJECT EXPERIENCE

HealthTech AI: Clinical Intelligence & Workflow Optimizer

- Engineered a Neurology AI platform using Gemini 2.5/2.0 Flash to auto-generate SOAP notes from consultation recordings, reducing neurologist consultation time by 48% (35→18 mins).
- Maintained ~96% speaker diarization accuracy using AssemblyAI and multi-model LLM fallback strategies; deployed containerized AWS services with Docker and Supabase while enforcing HIPAA-compliant PII governance and delivering 3 client demos.

Protein Interaction Network Analysis (NSCLC)

- Reduced protein interaction data from 4.9M to 184K records (~96% reduction), improving processing efficiency and enabling scalable graph-based analysis of NSCLC protein networks
- Analyzed 68 NSCLC-associated proteins using degree, closeness, betweenness, and eigenvector centrality, identifying TP53 as the strongest hub across all measures and ranking 10 key proteins for pathway-level insight extraction

CERTIFICATIONS

Microsoft Azure AI Fundamentals (AI-900) | Microsoft Azure Fundamentals (AZ-900) | Microsoft Azure Data Fundamentals (DP-900) | AWS Cloud Practitioner Essentials | Data Analysis Using Excel (Rice University) | Machine Learning: Linear Regression (NASSCOM FutureSkills)

SKILLS & INTERESTS

Technical Skills: Python (NetworkX, Scikit-learn, TensorFlow, Keras, XGBoost, OpenCV, DuckDB, Pandera, dbt-core), SQL, PySpark, R, PostgreSQL, Snowflake, Supabase, NoSQL Databases (MongoDB, Cassandra)

Data & Cloud: ETL/ELT Pipelines, Data Modeling, Star Schema, Data Warehousing, Data Transformation, Data Validation, Data Quality, Data Governance, HIPAA-Compliant PII Governance, Apache Airflow (workflow orchestration familiarity), AWS (S3), Azure (ADF, ADLS), GCP (BigQuery), Databricks, Spark, Hadoop, Hive

Analytics & ML: Regression Analysis, Classification, Clustering, Exploratory Data Analysis, Feature Engineering, Hypothesis Testing, A/B Testing, Transfer Learning, Computer Vision, Predictive Modeling, Tableau, Power BI, KPI Reporting

AI & Workflow: Gemini 2.5/2.0 Flash, AssemblyAI, Speaker Diarization, LLM Fallback Strategies, Git, GitHub, Docker, Jira, Microsoft Teams, Agile/Scrum, Cursor, VS Code, Excel, Jupyter

Domains: Healthcare Analytics, Supply Chain Analytics, Cybersecurity, Financial Analytics, Bioinformatics